

FINAL PROJECT REPORT

Online Learning Engagement and Student Performance Analysis

Team ID: MCA
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1. Introduction

1.1. Project overview

Online Learning Engagement and Student Performance Analysis is a Power BI data analytics project that uses student-level online education learning data to understand student engagement, academic performance, and dropout behavior. Online education platforms generate information about students' engagement, academic results, educational background, regional characteristics, and learning activity.

The project uses Power BI to convert the available student data into interactive visualizations. The analysis focuses on engagement level, total clicks, average score, performance, final results, risk level, dropout patterns, gender, region, and highest education where these are available in the project data and dashboard.

1.2. Objectives

- Understand overall student engagement and academic performance.
- Identify factors associated with student success and failure.
- Analyze student dropout patterns.
- Identify students or student groups that may be at risk of dropping out.
- Compare engagement levels across different student groups.
- Analyze the relationship between total clicks and academic performance.
- Examine differences in performance across regions and educational backgrounds.
- Monitor pass rates and dropout rates.
- Support data-driven student retention and intervention strategies.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

I am: an educational institution or online learning platform.

I'm trying to: understand students' online learning engagement, academic performance, and dropout behavior using student-level learning data.

But: it is difficult to identify the factors related to student success, failure, engagement, and dropout patterns from the available student data.

Because: the data contains information such as engagement level, total clicks, assessment

performance, final results, educational background, region, and risk level.

Which makes me feel: it is difficult to identify students or student groups that may need additional support and to make timely decisions for improving student performance and reducing dropout.

2.2. Project Proposal (Proposed Solution)

The proposed solution is an interactive Power BI dashboard and report for analyzing online learning engagement and student performance. The solution brings the available student-level information into one visual reporting environment so that performance, engagement, risk, dropout, and demographic patterns can be explored using charts, cards, and filters.

Project scope includes:

- Student engagement and learning activity analysis.
- Academic performance and final result analysis.
- Risk and dropout analysis.
- Comparison by gender, region, and highest education.
- Analysis of total clicks and engagement levels.
- Monitoring pass rates and dropout rates.

Basic System Requirements:

- Operating System: Windows 10 or Windows 11 (64-bit)
- Processor: Intel Core i3 / AMD Ryzen 3 or higher
- RAM: Minimum 8 GB
- Storage: At least 5 GB of free space
- Display: 1366 × 768 resolution or higher
- Database: MySQL / SQL Server or any compatible SQL database
- SQL Tool: MySQL Workbench / SQL Server Management Studio
- Power BI: Power BI Desktop (latest version recommended)

2.3. Initial Project Planning

The project is completed individually. The following product backlog summarizes the planned work.

Sprint	Functional Requirement	User Story Number	User Story / Task Story	Points	Priority
Sprint-1	Data Understanding	USN-1	As a project member, I can understand the online education dataset and identify important student fields.	2	High
Sprint-1	Data Cleaning	USN-2	As a project member, I can	2	High

			clean and prepare the student data for analysis in Power BI.		
Sprint-1	Data Preparation	USN-3	As a project member, I can prepare the required data fields and measures for the dashboard.	2	High
Sprint-2	Student Performance	USN-4	As a user, I can view student performance using average score and performance level.	2	High
Sprint-2	Student Engagement	USN-5	As a user, I can analyze student engagement using engagement level and average clicks.	2	High
Sprint-2	Final Results	USN-6	As a user, I can view Pass, Fail, Withdrawn, and Distinction results.	2	Medium
Sprint-3	Risk Analysis	USN-7	As a user, I can analyze student risk levels using the available risk categories.	2	High
Sprint-3	Dropout Analysis	USN-8	As a user, I can compare dropout rates across educational backgrounds.	2	High
Sprint-3	Demographic Analysis	USN-9	As a user, I can analyze student information by gender, region, and highest education.	1	Medium
Sprint-3	Dashboard Development	USN-10	As a user, I can explore student performance, engagement, risk, and dropout information in Power BI.	2	High

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

The main raw data source for the project is the file named `online_education_dataset.csv`. The dataset is used as the student-level source for the Power BI analysis. The project uses the available student information to study engagement, performance, final results, risk, dropout, demographic characteristics, and educational background. Exact dataset size, file size, source URL, and complete field list are not specified in the available project information and should be added if required.

Dataset Name: `Online_ Education _dataset.csv`

Dataset size: 32,593

Dataset format: CSV

3.2. Data Quality Report

The project uses student-level online education learning data. The dataset contains information required to study student engagement, academic performance, final results, risk, dropout behaviour, educational background, region, and other student characteristics.

Important fields used in the analysis include engagement level, total clicks, average score, performance level, final result, risk level, gender, region, and highest education.

3.3. Data Exploration and Preprocessing

The dataset was explored to understand student engagement, academic performance, final results, risk categories, dropout behavior, educational background, and regional characteristics. Relevant data was prepared for Power BI so that the information could be represented through measures, categories, filters, and visualizations.

The project analysis uses fields and categories that are visible in the dashboard, including engagement level, average clicks, average score, final result, risk level, gender, region, and highest education.

4. Data Visualization

4.1. Framing Business Questions

No.	Business Question	Visualization / Analysis
1	What is the distribution of total students across regions?	Bar chart showing total student count by region.
2	How are students distributed by final result?	Donut chart showing total students by final result
3	How does average total clicks vary by engagement level?	Column chart comparing average total clicks by engagement level
4	How does average score vary by engagement level?	Column chart comparing average score by engagement level.
5	How does average score vary by performance level?	Column chart comparing average score by performance level.
6	How are students distributed by performance level?	Donut chart showing total students by performance level.
7	How does Average Score vary across different regions?	Column chart comparing Average Score by region
8	How are students distributed according to their risk level?	Donut chart showing total students by risk level.

4.2. Developing Visualizations

The Power BI report uses visual elements to make the student data easier to understand. The developed analysis includes student totals, average score, engagement level, average clicks, final results, risk categories, dropout rate, pass rate, and demographic comparisons.

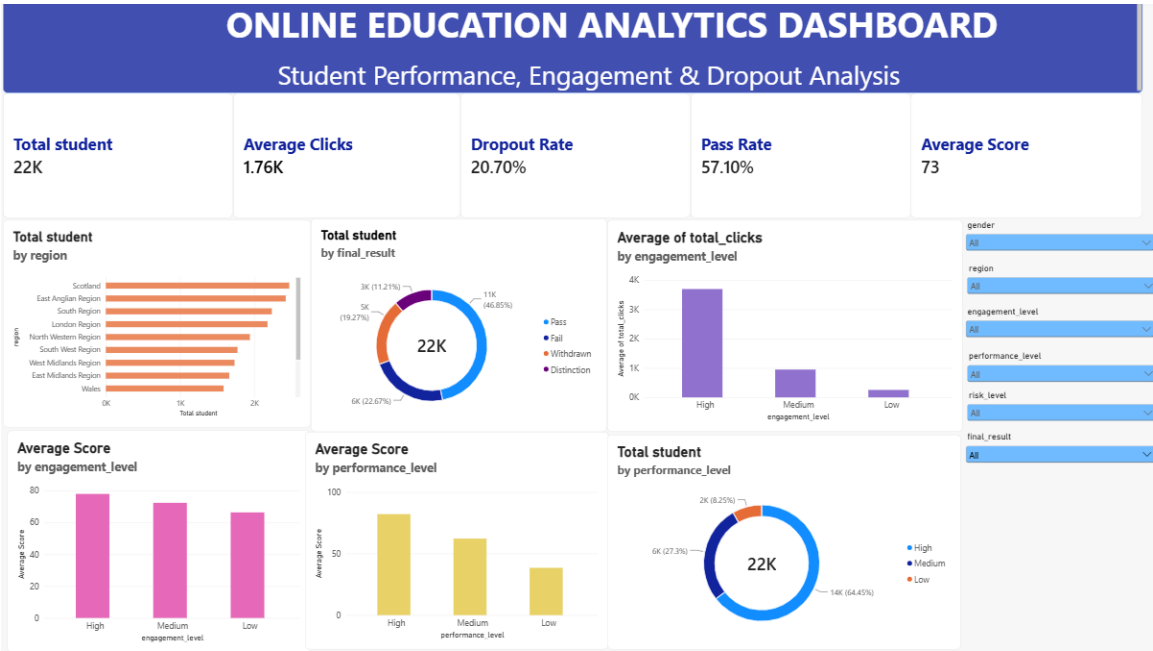
- Student count and summary information.
- Average score by region.
- Average clicks by engagement level.
- Final result distribution.
- Risk category analysis.
- Dropout rate by highest education.
- Pass rate by highest education.
- Gender and region based analysis.

5. Dashboard

5.1. Dashboard Design

Dashboard 1: Online Education Analytics Dashboard

The Online Education Analytics Dashboard provides an overview of student performance, engagement, and dropout patterns. It shows 22K total students, 1.76K average clicks, a 20.7% dropout rate, a 57.10% pass rate, and an average score of 73. The dashboard indicates that students with high engagement have more clicks and higher average scores compared with medium- and low-engagement students. The performance chart shows that medium-performance students form the largest group, followed by high- and low-performance students. The dashboard also allows users to analyze student data using filters such as gender, region, engagement level, performance level, risk level, and final result, making it easier to understand different student groups and identify important patterns.

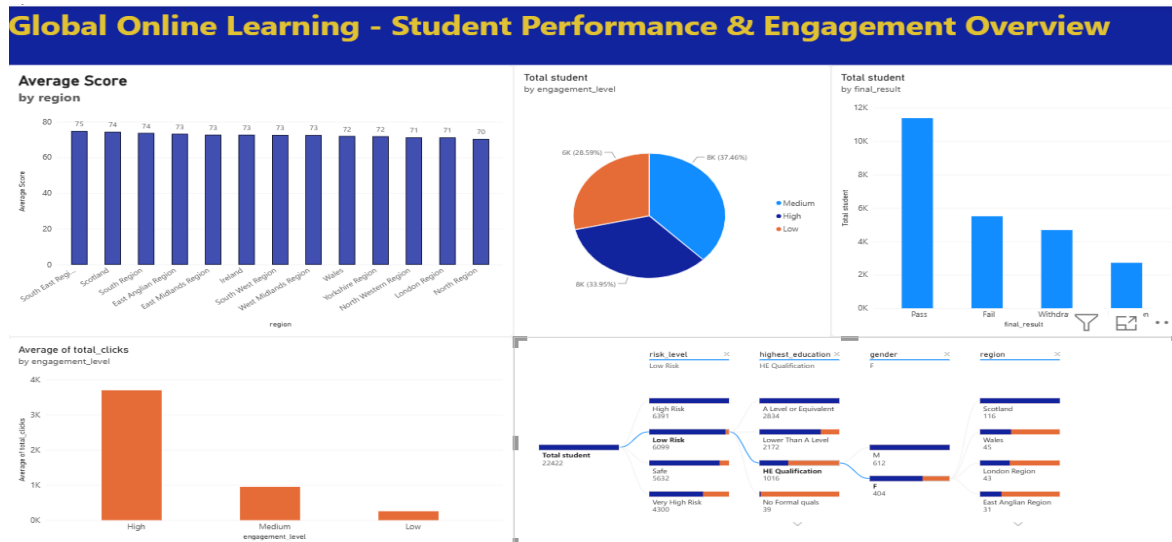


6. Report

6.1. Report Design

The Power BI report combines the developed visualizations into an interactive report. Available filters include Gender, Region, Engagement Level, Performance Level, Risk Level, and Final Result. These filters allow the user to explore different student growth

Global Online Learning – Student Performance & Engagement Overview



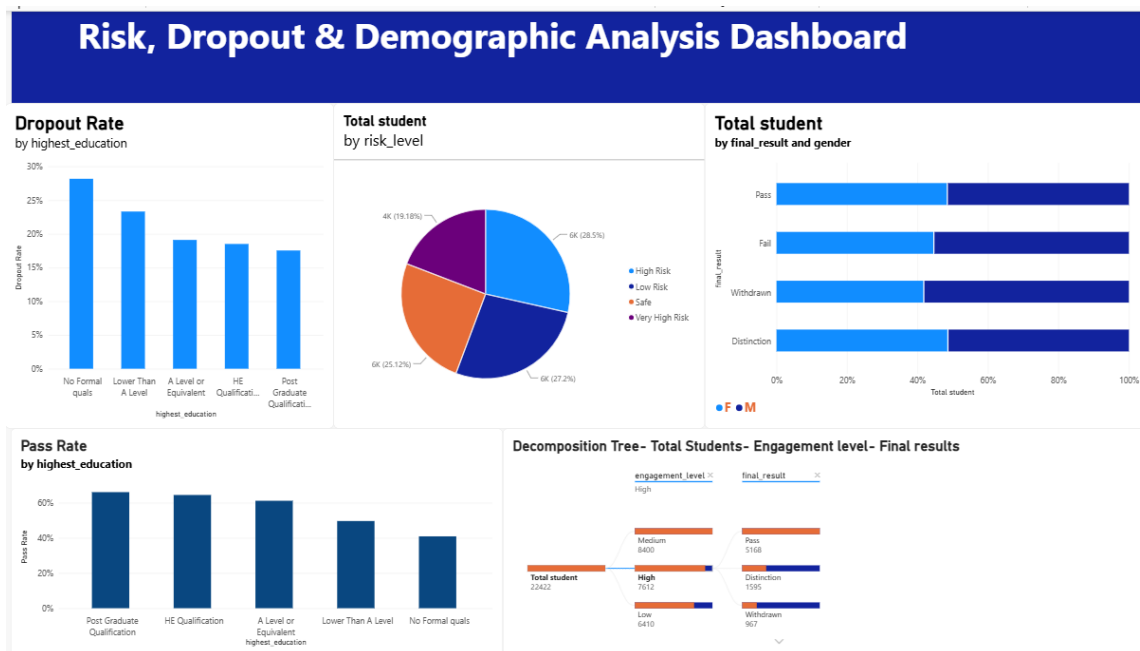
The first dashboard provides an overview of student performance and engagement. It contains student totals, average score, engagement levels, average clicks, final results, and regional analysis.

The dashboard shows a total of 22,422 students. Engagement is grouped into High, Medium, and Low. Average clicks are highest for the High engagement group, followed by Medium and Low. Final results include Pass, Fail, Withdrawn, and Distinction.

6.2: Risk, Dropout & Demographic Analysis Dashboard

The second dashboard focuses on risk, dropout, and demographic analysis. It includes risk categories, dropout rate by highest education, pass rate by highest education, and demographic information.

The dashboard shows that dropout rates differ across highest-education categories. The visible analysis shows the highest dropout rate for No Formal qualifications and the lowest visible dropout rate for Post Graduate Qualification. Pass rate is highest for Post Graduate Qualification and lowest for No Formal qualifications in the displayed analysis.



7. Performance Testing

7.1. Utilization of Data Filters

Filter / Slicer	Purpose
Gender	Compare outcomes by gender
Region	Focus the analysis on specific regions
Engagement Level	Filter High, Medium, or Low engagement students
Performance Level	Filter High, Medium, or Low performance students
Risk Level	Filter Low Risk, Safe, High Risk, or Very High Risk
Final Result	Filter Pass, Fail, Withdrawn, or Distinction

7.2. Number of Calculation Columns / Measures

Five core report measures are identifiable in the supplied PBIX report definition: Total Students, Average Score, Average Clicks, Dropout Rate, and Pass Rate. The preprocessing documentation additionally states that a unique index column was added to identify each record. A separate calculation-column count is not explicitly exposed in the supplied report documentation, so no additional number is asserted here.

7.3. Number of Visualizations

Visual Element	Count
Card visuals / KPI cards	10
Slicers	6
Column charts	3
Donut chart	1
Pie chart	1
Text boxes	3
Total visuals across supplied PBIX definition	24

8. Conclusion/Observation

The project demonstrates how Power BI can be used to analyze online learning engagement and student performance from student-level learning data. The analysis brings together engagement level, average clicks, average score, final results, risk level, dropout rate, pass rate, region, gender, and highest education.

The dashboard makes it easier to compare student groups and observe patterns in engagement, academic performance, risk, and dropout behavior. The available results can support educational institutions and online learning platforms in understanding student outcomes and planning data-driven retention and intervention strategies.

9. Future Scope

- Add more student activity and learning behavior data.
- Include time-based analysis of engagement and academic performance.
- Add predictive analysis to identify future dropout risk.
- Provide more detailed drill-down analysis for student groups.
- Connect the dashboard to regularly updated data sources.
- Add additional Power BI measures and interactive visuals as required.
- Improve reporting features for monitoring student retention.

10. Appendix

10.1. Source Code (if any)

The project is developed using Power BI. Power Query transformations, calculated columns, measures, or other technical code can be added here if require

10.2. GitHub Link

GitHub Link: <https://github.com/kambhampatimounika/online-learning-engagement-and-student-performance-anlaysis>